Applying the Spectral Scale Selection Pipeline to Autoregressive Processes

Detailed elaboration and possible research directions

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May 19, 2026

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1 Executive summary

Your notes propose to replace the ordinary scale grid

$$\varepsilon_1 < \cdots < \varepsilon_N$$

by the embedding-dimension grid

$$1, 2, \ldots, k_{\max}$$

for sliding-window point clouds of an autoregressive process. The aim is to construct a triple

$$(f, \{K_k\}_{k=1}^{k_{\max}}, \oplus)$$

that is *spectrally optimizable*, and to ask whether the principal Perron–Frobenius eigenvector selects the true autoregressive order. This is an interesting idea, but it has one serious trap: if the complexes are nested only because the construction forces

$$K_1 \subseteq K_2 \subseteq \cdots \subseteq K_{\max},$$

then the spectral method may select a feature of the artificial filtration rather than a feature of the stochastic dynamics.

The central recommendation of this report is therefore:

Principle 1 (Do not let k only be a formal filtration index)**.** *The dimension k should enter the construction through the geometry, prediction structure, residual structure, persistence structure, or temporal dependence structure of the AR process. A purely combinatorial complex that adds all consecutive j-simplices for $j \leq k-1$ is useful as a temporal scaffold, but by itself it cannot recover the AR order because it does not use the values of the observations.*

I recommend developing the AR application around three complementary pipelines.

Pipeline A. Hyperparameter pipeline. For each k, build a geometric filtration $K_\varepsilon(\mathcal{P}_k)$ and run your existing scale-selection method in ε. Then study the curve

$$k \longmapsto \varepsilon_{\text{crit}}(k)$$

and/or a stability score. This is the safest route for a master's thesis because it reuses your existing method and turns AR order selection into a detectable change point in k.

Pipeline B. Common-space pipeline. Lift all delay clouds into a common space $\mathbb{R}^{k_{\max}}$, build comparable complexes on a common vertex set, and use a similarity matrix indexed by k. This is closest to your notes. It is promising if the lift is chosen carefully, for example by conditional-mean prediction or by a Hankel/SVD state-space lift.

Pipeline C. Persistence-kernel pipeline. For each k, compute persistence diagrams D_k and form

$$M_{k\ell} = \phi(d(D_k, D_\ell)) \quad \text{or} \quad M_{k\ell} = \kappa(D_k, D_\ell),$$

where d is bottleneck or Wasserstein distance, or κ is a positive-definite persistence kernel. Then apply the spectral step to M. This route has the strongest existing literature support because persistence stability and diagram kernels are well studied.

The most realistic mathematical target is not the exact statement "$\arg\max_k v_k^* = p$" in full generality. A more plausible and defensible target is:

> For a stationary AR(p) process, the sequence of geometric, residual, or persistence summaries changes substantially until k reaches roughly $p+1$ and stabilizes afterwards. The spectral method should detect this transition, so the selected k should estimate either p, $p+1$, or the first stable embedding dimension depending on the chosen convention.

This convention matters. The AR equation

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + Z_t$$

relates $p+1$ consecutive observations. Hence a geometric hyperplane relation first appears naturally in dimension $p+1$, whereas prediction of X_t from the past requires p lags. Your method may therefore select p or $p+1$ depending on whether k means "number of past predictors" or "number of coordinates in the delay vector."

2 Background: AR processes and sliding-window geometry

Let $(X_t)_{t\in\mathbb{Z}}$ be a stationary AR(p) process

$$X_t = \sum_{i=1}^{p} \phi_i X_{t-i} + Z_t, \tag{1}$$

where (Z_t) is an innovation sequence, usually assumed to be independent and centered. The classical time-series literature treats (1) as a finite-memory linear stochastic model; see Brockwell and Davis [3]. Your TDA-based idea instead studies the geometry of the delay vectors

$$Y_t^{(k)} = (X_t, X_{t-1}, \ldots, X_{t-k+1}) \in \mathbb{R}^k, \qquad \mathcal{P}_k = \{Y_t^{(k)} : t = k, \ldots, T\} \subset \mathbb{R}^k. \tag{2}$$

This is structurally close to delay-coordinate methods in dynamical systems. Takens' theorem justifies delay embeddings for generic deterministic dynamical systems [12]; Sauer, Yorke, and Casdagli extended this viewpoint in the influential "embedology" framework [11]. For TDA, Perea and Harer developed a systematic theory of sliding windows and persistent homology for signal analysis [9]. These papers do not solve AR order estimation, but they provide good precedence for the central operation: turning a one-dimensional observed process into point clouds indexed by a window length.

2.1 Why the number $p+1$ appears geometrically

Suppose first that we use forward coordinates

$$\widetilde{Y}_t^{(k)} = (X_t, X_{t+1}, \ldots, X_{t+k-1}).$$

For $k = p+1$, equation (1) gives

$$X_{t+p} - \phi_1 X_{t+p-1} - \cdots - \phi_p X_t = Z_{t+p}. \tag{3}$$

Hence $\widetilde{Y}_t^{(p+1)}$ lies near the affine hyperplane

$$H_0 = \{(x_0, \ldots, x_p) : x_p - \phi_1 x_{p-1} - \cdots - \phi_p x_0 = 0\}$$

when the innovation variance is small. If the innovations are bounded, for example $Z_t \in [-a, a]$, the point cloud lies in the slab

$$\{x \in \mathbb{R}^{p+1} : |x_p - \phi_1 x_{p-1} - \cdots - \phi_p x_0| \leq a\}.$$

For $k > p + 1$, one obtains several shifted relations:

$$|x_j - \phi_1 x_{j-1} - \cdots - \phi_p x_{j-p}| leq a, \qquad j = p, \ldots, k-1. \tag{4}$$

Thus the support is concentrated near the intersection of $k - p$ slabs, and in the noiseless case lies in a p-dimensional linear subspace. This explains why the intrinsic dimension should stop increasing once the delay dimension is beyond p or $p + 1$, depending on the precise indexing convention.

Remark 1 (Backward versus forward windows)**.** *For stationary processes, the distributional geometry of backward and forward windows is essentially the same up to coordinate reversal, provided one compares corresponding vectors. What changes is the interpretation: backward windows are natural for prediction, while forward windows make the hyperplane relation* (3) *visually immediate.*

2.2 Support as a zonoid for bounded innovations

If the AR process is causal and stationary, it has a moving-average representation

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}, \tag{5}$$

where $\psi_0 = 1$ and the coefficients satisfy the recursion

$$\psi_j = \phi_1 \psi_{j-1} + \cdots + \phi_p \psi_{j-p}, \qquad \psi_j = 0 \text{ for } j < 0.$$

If $Z_t \in [-1, 1]$, then the support of a delay vector can be written as a Minkowski sum of line segments. For example, in dimension $p + 1$,

$$Y_t^{(p+1)} \in \sum_{j=0}^{\infty} g_j[-1, 1], \qquad g_j = (\psi_j, \psi_{j-1}, \ldots, \psi_{j-p})^\top, \tag{6}$$

with the convention $\psi_j = 0$ for $j < 0$. Finite truncations

$$Z^{(m)} = \sum_{j=0}^{m} g_j[-1, 1]$$

are zonotopes. This observation is valuable because it turns the stochastic time-series model into a convex-geometric object. A possible thesis contribution would be to numerically test whether your spectral scale selection detects the dimension at which these zonotopal supports stop changing qualitatively.

For Gaussian innovations the support is all of $\mathbb{R}^k$, so the slab/zonoid statement must be replaced by a high-probability concentration statement. The cloud is not literally bounded, but it is concentrated near the AR subspace when the innovation variance is small relative to the stationary variance.

Meaning of k	First meaningful AR dimension	Expected selected value
k is the number of predictors $(X_{t-1}, \dots, X_{t-k})$ used to predict X_t.	$k = p$ predictors are sufficient.	p
k is the number of coordinates in the delay vector $(X_t, \dots, X_{t-k+1})$.	The first direct linear relation involves $p+1$ coordinates.	$p+1$
k is a temporal simplex dimension parameter.	K_k contains simplices of dimension $k-1$ by construction.	Not intrinsically tied to p unless geometry enters.

Table 1: Three meanings of k and the resulting interpretation of the selected index.

3 What exactly should be selected: p or $p+1$?

There are two natural conventions.

For your thesis, I recommend stating the target as follows:

$$\widehat{p} = k_{\text{crit}} - 1$$

when k indexes delay-vector dimension, and

$$\widehat{p} = k_{\text{crit}}$$

when k indexes the number of lagged predictors. This avoids an artificial off-by-one ambiguity.

4 Dimension lifting: ways to compare $\mathcal{P}_k \subset \mathbb{R}^k$ across k

Your notes propose a prediction-based lift

$$p_t^k := (X_t, X_{t-1}, \dots, X_{t-k+1}, \widehat{X}_{t-k}, \dots, \widehat{X}_{t-k_{\max}+1}) \in \mathbb{R}^{k_{\max}}.$$

This is a strong idea because it makes every point cloud live in the same ambient space and on the same vertex set

$$V = \{k_{\max}, \dots, T\}.$$

Below are several possible lifts, ordered roughly from simplest to most promising.

4.1 Lift 1: zero padding or mean padding

Define

$$L_k(Y_t^{(k)}) = (X_t, \dots, X_{t-k+1}, 0, \dots, 0) \in \mathbb{R}^{k_{\max}}.$$

This is easy but usually not recommended. The appended zero coordinates are not stochastic predictions; they create artificial axes and artificial distances. Mean padding,

$$L_k(Y_t^{(k)}) = (X_t, \dots, X_{t-k+1}, \bar{X}, \dots, \bar{X}),$$

is slightly better for centered data but still arbitrary.

Warning 1. *Zero padding is useful only as a baseline. If the method works only with zero padding, it may be detecting the padding rule rather than the AR order.*

4.2 Lift 2: conditional-mean prediction lift

This is your proposed lift. Let

$$q_t^k = (X_t, dots, X_{t-k+1}) \in \mathbb{R}^k, \qquad z_t^k = (X_{t-k}, \dots, X_{t-k_{\max}+1}) \in \mathbb{R}^{k_{\max}-k}.$$

Estimate

$$(\widehat{A}_k, \widehat{b}_k) \in \arg\min_{A,b} \sum_{r=k_{\max}}^{T} \|z_r^k - Aq_r^k - b\|^2, \tag{7}$$

where $A \in \mathbb{R}^{(k_{\max}-k)\times k}$ and $b \in \mathbb{R}^{k_{\max}-k}$. Then set

$$\widehat{z}_t^k = \widehat{A}_k q_t^k + \widehat{b}_k, \qquad p_t^k = (q_t^k, \widehat{z}_t^k).$$

This is attractive because for Gaussian stationary AR processes the linear predictor is the conditional expectation. It therefore has a statistical interpretation: $\mathcal{Q}_k$ is the cloud obtained by filling the missing older lags with their best linear predictions from the observed k lags.

Expected AR behavior. For $k < p$, the future/older coordinates are under-predicted because the window misses relevant state information. For $k \geq p$, prediction error should stabilize because the Markov state of the AR process has been captured. Therefore the geometry of $\mathcal{Q}_k$ should change strongly up to $k = p$ and weakly afterwards.

4.3 Lift 3: prediction lift with residual coordinates

A small modification may be even better. Instead of replacing the missing coordinates by predictions only, append a residual-quality coordinate:

$$r_t^k = \|z_t^k - \widehat{A}_k q_t^k - \widehat{b}_k\|.$$

One can define

$$\widetilde{p}_t^k = (q_t^k, \widehat{z}_t^k, \lambda r_t^k) \in \mathbb{R}^{k_{\max}+1}$$

with a tuning parameter $\lambda \geq 0$. This makes the lift sensitive to the fact that $k < p$ gives unstable predictions, whereas $k \geq p$ should reduce residual variation. It also connects the method with classical order selection: AR order selection is fundamentally about when additional lags stop improving prediction.

A functional can then explicitly use residuals:

$$f_{\text{res}}(K_k) = g(K_k) \cdot h(\widehat{\sigma}_k^2), \qquad \widehat{\sigma}_k^2 = \frac{1}{T - k_{\max} + 1} \sum_{t=k_{\max}}^{T} (r_t^k)^2,$$

where g is topological/combinatorial and h penalizes large residual variance.

4.4 Lift 4: Hankel/SVD lift

Form the Hankel matrix

$$H_{k_{\max}} = \begin{pmatrix} X_{k_{\max}} & X_{k_{\max}+1} & \cdots & X_T \\ X_{k_{\max}-1} & X_{k_{\max}} & \cdots & X_{T-1} \\ \vdots & \vdots & \ddots & \vdots \\ X_1 & X_2 & \cdots & X_{T-k_{\max}+1} \end{pmatrix}.$$

For each k, one may regard the first k rows as observed and reconstruct the remaining rows through a low-rank approximation. This connects the construction with singular-spectrum analysis and state-space ideas. For a low-noise AR process, the leading singular directions should stabilize once the state dimension is captured. The benefit is that the lift is global and data-adaptive; the drawback is that it is less directly tied to your existing simplicial-complex pipeline.

4.5 Lift 5: masked metric without imputation

Instead of imputing missing coordinates, define a distance between two partial delay vectors by using only the shared coordinates and normalizing by the number of shared coordinates:

$$d_{k\ell}(t,s)^2 = \frac{k_{\max}}{\min(k,\ell)} \sum_{i=0}^{\min(k,\ell)-1} (X_{t-i} - X_{s-i})^2.$$

This yields cross-k distances without forcing all points into a common Euclidean cloud. It is useful for diagram-kernel or graph-based methods. It is less compatible with ordinary Vietoris–Rips filtrations on a single common point cloud, but it avoids the modeling assumptions of imputation.

5 Constructing complexes without losing temporal information

Your notes define a complex

$$K_k = \langle \{\{r, \dots, r+j\} : 0 \le j \le k-1,\ r = k_{\max}, \dots, T-j\} \rangle\,. \tag{8}$$

This is a valid filtration

$$K_1 \subseteq K_2 \subseteq \cdots \subseteq K_{k_{\max}}.$$

It preserves temporal contiguity because all simplices are generated by consecutive time intervals. However, it has no dependence on the values X_t. Hence it cannot by itself recover p.

Warning 2 (Pure temporal complexes are too deterministic)**.** *For fixed T and $k_{\max}$, the complex (8) is the same for every time series. It is therefore a temporal scaffold, not yet a data-dependent AR-order detector.*

The next constructions combine your temporal scaffold with the geometry of the lifted clouds $\mathcal{Q}_k$.

5.1 Construction A: temporal Vietoris–Rips complex

For $\varepsilon > 0$ and a temporal horizon $m \in \mathbb{N}$, define a graph on V by

$$(t,s) \in E_{k,\varepsilon,m} \quad \Longleftrightarrow \quad \|p_t^k - p_s^k\| \le \varepsilon \quad \text{and} \quad |t-s| \le m.$$

Then let $K_{k,\varepsilon,m}$ be the flag complex of this graph. The parameter m prevents the complex from ignoring time order completely; the parameter ε preserves the geometric TDA interpretation.

There are two useful regimes:

1. Fix $m = k$ and run scale selection in ε for each k.
2. Fix ε by your spectral scale selection and then study the dimension index k.

5.2 Construction B: intersection of temporal and geometric complexes

Let T_k denote your pure temporal complex (8). Let $G_{k,\varepsilon} = \mathrm{VR}_\varepsilon(\mathcal{Q}_k)$. Define

$$K_{k,\varepsilon} = T_k \cap G_{k,\varepsilon}.$$

A simplex is included only if it is temporally admissible and geometrically close. This is a clean way to preserve time without losing the metric information.

The operation $\cap$ has an intuitive interpretation: the temporal complex proposes allowed local time blocks, and the geometric complex accepts only those blocks whose delay vectors are mutually close. This construction is also computationally attractive because T_k is sparse.

5.3 Construction C: union of temporal and geometric evidence

Alternatively define

$$K_{k,\varepsilon} = T_k \cup G_{k,\varepsilon}.$$

This keeps all temporal simplices and adds geometric recurrence simplices. It is useful if one wants the complex to remain connected along the observed trajectory. However, it may wash out topological differences because the temporal backbone is always present.

5.4 Construction D: recurrence complexes

A recurrence plot connects times t and s if the corresponding states are close. In your notation this gives the graph

$$R_{k,\varepsilon} = \{(t, s) : \|Y_t^{(k)} - Y_s^{(k)}\| \le \varepsilon\}.$$

Taking the flag complex of this graph produces a recurrence complex. This construction is well aligned with delay-coordinate methods: periodic or quasiperiodic series generate recurrent loops, while AR processes generate clouds whose recurrence structure depends on the covariance and innovation distribution. It is a good candidate for the hyperparameter pipeline.

5.5 Construction E: directed path complex

If direction of time is essential, use a directed graph

$$t \to s \quad \Longleftrightarrow \quad 0 < s - t \le m \quad \text{and} \quad \|p_t^k - p_s^k\| \le \varepsilon.$$

Instead of taking an undirected flag complex, one can study a path complex or directed flag complex. Path homology of digraphs was introduced by Grigor'yan, Lin, Muranov, and Yau [7]. This is more sophisticated, but it matches your phrase "ordered temporal simplex" better than ordinary simplicial homology. A master's-thesis-safe compromise is to construct the directed graph but then compare both its undirected flag homology and directed graph summaries such as in/out-degree imbalance, directed cycles, and transition entropy.

6 Operations $\oplus$ on complexes

Your general scalar product has the form

$$\langle K, L\rangle_f = f(K) + f(L) - f(K \oplus L), \tag{9}$$

with $\oplus$ often chosen as union. In a filtration, $K_i \cup K_j = K_{\max(i,j)}$ and $K_i \cap K_j = K_{\min(i,j)}$, so the matrix becomes a meet/min-type matrix whenever the functional is evaluated on the smaller complex. Across different k values, however, the complexes need not be nested. The operation therefore becomes a modeling choice.

6.1 Union

The union

$$K_k \oplus K_\ell = K_k \cup K_\ell$$

asks: how much additional structure is needed to cover both dimension choices? If f is subadditive, then (9) is nonnegative. For example, if f is an LS-category-like complexity, then a large value of

$$f(K_k) + f(K_\ell) - f(K_k \cup K_\ell)$$

means that K_k and K_ℓ share explanatory structure.

6.2 Intersection

The intersection

$$K_k \oplus K_\ell = K_k \cap K_\ell$$

is useful when one wants common robust structure. However, using intersection inside (9) no longer has the same subadditivity interpretation. A direct similarity

$$S_{k\ell} = f(K_k \cap K_\ell)$$

may be better.

6.3 Join and temporal concatenation

The join $K_k * K_\ell$ is usually too large and hard to interpret for this problem. More promising is a temporal concatenation operation: combine a simplex from K_k and a later simplex from K_ℓ if their time intervals are ordered. This produces a complex that explicitly represents transitions from a k-memory description to an ℓ-memory description. This is novel but more difficult, so I would present it as an open research direction rather than the main thesis method.

6.4 Diagram-level operations

Instead of operating on complexes, compute persistence diagrams D_k and define

$$S_{k\ell} = \exp\left(-\frac{d(D_k, D_\ell)^2}{2\sigma^2}\right),$$

or use a positive-definite persistence kernel. Reininghaus et al. introduced a stable multi-scale kernel for persistence diagrams [10], and Carriere et al. introduced a sliced Wasserstein kernel [6]. This is likely the cleanest way to compare non-nested complexes indexed by k.

7 Functionals f: what should be measured?

The functional is the most important choice. A monotone functional usually makes the Perron eigenvector monotone or near-monotone, so it tends to select an endpoint rather than a meaningful transition. The useful functionals are those that increase and then decrease, or decrease and then plateau, or measure a contrast between topology and combinatorics.

7.1 Class 1: topology-combinatorics products

Your thesis already considers functionals of the form

$$f(K) = f_{\text{top}}(K)\, f_{\text{comb}}(K).$$

For AR clouds, good first candidates are

$$\begin{aligned}
f_1(K) &= \beta_0(K)\, f_1(K), \\
f_2(K) &= \beta_1(K)\, f_1(K), \\
f_3(K) &= (\beta_0(K) + \alpha\beta_1(K))(f_1(K) + \gamma f_2(K)), \\
f_4(K) &= \left(\sum_{j\geq 0} w_j \beta_j(K)\right)\left(\sum_{j\geq 0} u_j f_j(K)\right).
\end{aligned}$$

Here $f_j(K)$ denotes the number of j-simplices. The product $\beta_0 f_1$ is useful because f_1 grows as edges appear, while β_0 decreases as components merge. This creates a peak near a structural transition.

7.2 Class 2: persistence summaries

For each k, let $D_k^{(q)}$ be the q-dimensional persistence diagram of a filtration built from $\mathcal{P}_k$ or $\mathcal{Q}_k$. Define

$$\begin{aligned}
\text{Pers}_q^{(r)}(k) &= \sum_{(b,d)\in D_k^{(q)}} (d-b)^r, \\
\text{MaxPers}_q(k) &= \max_{(b,d)\in D_k^{(q)}} (d-b), \\
\text{Entropy}_q(k) &= -\sum_i \rho_i \log \rho_i, \qquad \rho_i = \frac{d_i - b_i}{\sum_j (d_j - b_j)}.
\end{aligned}$$

Perea and Harer showed that persistent homology of sliding-window embeddings is informative for periodicity and signal structure [9]. For AR processes, these summaries may not produce large loops unless the AR coefficients are oscillatory or near complex roots. Therefore persistence summaries should be interpreted as geometry-change detectors rather than direct "hole counters."

7.3 Class 3: residual and prediction functionals

Because AR order is a prediction-memory concept, include prediction quality:

$$\widehat{\sigma}_k^2 = \frac{1}{T - k_{\max} + 1} \sum_{t=k_{\max}}^{T} \|z_t^k - \widehat{A}_k q_t^k - \widehat{b}_k\|^2.$$

Possible functionals are

$$f_{\mathrm{pred},1}(k) = \frac{g(K_k)}{\widehat{\sigma}_k^2 + \eta},$$
$$f_{\mathrm{pred},2}(k) = g(K_k)\left(\widehat{\sigma}_{k-1}^2 - \widehat{\sigma}_k^2\right)_+,$$
$$f_{\mathrm{pred},3}(k) = g(K_k) \cdot \frac{\widehat{\sigma}_{k-1}^2 - \widehat{\sigma}_k^2}{\widehat{\sigma}_{k-1}^2 + \eta}.$$

These are promising because they can peak at the final substantial improvement in prediction, which should occur around p.

7.4 Class 4: intrinsic-dimension functionals

Let $\lambda_1^{(k)} \geq \cdots \geq \lambda_k^{(k)}$ be the eigenvalues of the sample covariance matrix of $\mathcal{P}_k$. Define

$$d_{\mathrm{eff}}(k) = \frac{\left(\sum_i \lambda_i^{(k)}\right)^2}{\sum_i (\lambda_i^{(k)})^2}$$

or the participation ratio. Another score is the spectral gap

$$\mathrm{gap}(k) = \frac{\lambda_p^{(k)}}{\lambda_{p+1}^{(k)} + \eta},$$

although this depends on p and is therefore not directly usable for estimation. A data-driven alternative is the largest covariance eigengap

$$G(k) = \max_{1 \leq j < k} \log \frac{\lambda_j^{(k)} + \eta}{\lambda_{j+1}^{(k)} + \eta}.$$

Intrinsic-dimension scores are not purely topological, but they are very relevant because the AR geometry predicts dimension saturation.

7.5 Class 5: temporal graph functionals

For a temporal recurrence graph $R_{k,\varepsilon}$, use

$$f_{\mathrm{rec},1}(k) = \#\{(t,s) : \|Y_t^{(k)} - Y_s^{(k)}\| \leq \varepsilon\},$$
$$f_{\mathrm{rec},2}(k) = \sum_t \deg(t)^2,$$
$$f_{\mathrm{rec},3}(k) = \text{number of triangles in } R_{k,\varepsilon},$$
$$f_{\mathrm{rec},4}(k) = \text{clustering coefficient of } R_{k,\varepsilon}.$$

These are computationally cheap and can be combined with Betti numbers. They are useful baselines before computing full persistent homology.

8 Scale similarity matrices indexed by k

There are several possible matrices.

8.1 Matrix A: meet matrix from a true filtration

If you genuinely have

$$K_1 \subseteq K_2 \subseteq \cdots \subseteq K_{k_{\max}},$$

then define

$$\Sigma_{k\ell} = f(K_{k\wedge\ell}).$$

This is exactly your min/meet-matrix setting. The drawback is that a forced filtration may encode only constructional monotonicity.

8.2 Matrix B: union-overlap matrix

For non-nested complexes on a common vertex set,

$$\Sigma_{k\ell} = f(K_k) + f(K_\ell) - f(K_k \cup K_\ell).$$

This works best when f is subadditive. If f is not subadditive, the entries may become negative; in that case use a shifted version

$$\widetilde{\Sigma}_{k\ell} = \Sigma_{k\ell} - \min_{a,b} \Sigma_{ab} + \eta$$

with $\eta > 0$ before applying Perron–Frobenius.

8.3 Matrix C: diagram distance matrix

Compute diagrams D_k and set

$$\Sigma_{k\ell} = \exp\left(-\frac{W_q(D_k, D_\ell)^2}{2\sigma^2}\right).$$

This is simple and stable in practice, although Gaussian transforms of arbitrary diagram distances are not automatically positive definite. For rigorous positive-definite kernels, use established persistence kernels such as Reininghaus et al. [10] or Carriere et al. [6]. Persistence landscapes [4] and persistence images [1] also give vector representations from which ordinary kernels can be built.

8.4 Matrix D: change-point matrix

If the goal is to find the first stable k, construct a matrix from adjacent differences:

$$\Delta_k = d(S_k, S_{k+1}),$$

where S_k is a summary of the complex or diagram at dimension k. Then define

$$\Sigma_{k\ell} = \exp\left(-\frac{|\Delta_k - \Delta_\ell|^2}{2\sigma^2}\right)$$

or run your spectral selection on the cumulative contrast

$$c_k = \Delta_k \cdot \left(\sum_{j>k} \Delta_j + \eta\right)^{-1}.$$

This directly targets an elbow: large change before p, small change after p.

9 Can the Perron eigenvector recover the AR order?

The exact claim

$$\arg \max_{k=1,\dots,k_{\max}} v_k^* = p$$

is too strong without assumptions on the functional, innovation distribution, AR coefficients, sample size, and filtration scale. Some AR processes are geometrically hard to distinguish: for example, an AR(p) with very small high-order coefficient ϕ_p may look like a lower-order process at finite sample size. Also, Gaussian innovations make all delay clouds full-dimensional in support, so only covariance concentration or prediction improvement can reveal the order.

A more plausible theorem would be a consistency statement for a contrast functional.

Conjecture 1 (Prediction-topology order consistency)**.** *Let (X_t) be a stationary causal AR(p) process with nondegenerate innovations and sufficiently separated coefficients, and let $k_{\max} > p$. Suppose $K_{k,\varepsilon}$ is built from a prediction lift and a temporal-geometric Rips construction. Let*

$$c_k = g(K_{k,\varepsilon_k}) \cdot \frac{\widehat{\sigma}_{k-1}^2 - \widehat{\sigma}_k^2}{\widehat{\sigma}_{k-1}^2 + \eta},$$

where ε_k is selected by the internal scale-selection pipeline and g is a bounded, stable topological-combinatorial functional. Then, under appropriate sampling and separation assumptions, the dominant peak of c_k occurs at $k = p$, and the Perron eigenvector of the meet matrix generated by c_k selects p with probability tending to one as $T \to \infty$.

This conjecture is much more promising than a theorem based only on the temporal simplex filtration. It combines the statistical fact "prediction improves until the correct AR order" with your spectral regularization of scale profiles.

10 Concrete recommendations for the thesis

10.1 Recommended main route

Use the hyperparameter pipeline as the main empirical thesis contribution:

1. Simulate AR(p) processes for $p = 1, 2, 3, 4$ with stable coefficients.
2. For each $k = 1, \dots, k_{\max}$, build the delay cloud $\mathcal{P}_k$.
3. For each k, run a Vietoris–Rips filtration over ε.
4. Compute several functionals, for example $\beta_0 f_1$, total persistence, number of edges times number of connected components, and residual-weighted variants.
5. Apply your spectral scale selection in ε to obtain $\varepsilon_{\mathrm{crit}}(k)$ and a selected value $v_\varepsilon^*(k)$.
6. Study the curves
$$k \mapsto \varepsilon_{\mathrm{crit}}(k), \qquad k \mapsto f(K_{k,\varepsilon_{\mathrm{crit}}(k)}), \qquad k \mapsto \widehat{\sigma}_k^2.$$
7. Detect a change point or peak in k using your spectral method again.

This approach lets you honestly say: "I do not replace ε by k immediately; I first use the original scale-selection method inside each embedding dimension, and then use the resulting scale profile to infer the AR order."

10.2 Recommended secondary route

Develop your common-space lift as a second experiment:

1. Use the regression lift (7).
2. Build $\mathcal{Q}_k \subset \mathbb{R}^{k_{\max}}$ on the common vertex set V.
3. Construct $K_{k,\varepsilon} = T_k \cap \mathrm{VR}_\varepsilon(\mathcal{Q}_k)$.
4. Either fix ε globally or choose ε_k using the internal pipeline.
5. Build a similarity matrix across k using union overlap or diagram kernels.
6. Compare the selected k with the true p and $p+1$.

This route is closer to your original notes and can become the novel part of the thesis.

10.3 Ideas to explicitly mark as exploratory

The following ideas are interesting but should probably not be the backbone of the thesis unless the numerical experiments are extremely convincing:

- directed path homology of temporal recurrence graphs;
- temporal concatenation operations on complexes;
- full zigzag persistence over non-nested K_k;
- rigorous recovery theorem for all stationary AR(p) processes;
- LS-category-based functionals, unless computationally feasible for your complexes.

Zigzag persistence is relevant because it handles sequences of spaces connected by maps in both directions, not only nested filtrations [5]. It is mathematically elegant, but implementation and interpretation may become too large for the available scope.

11 Possible numerical experiment design

11.1 Data-generating models

Use the following families.

(i) **AR(1):** $X_t = \phi X_{t-1} + Z_t$, with $\phi \in \{0.2, 0.6, 0.9, -0.6\}$.

(ii) **AR(2) oscillatory:** choose roots close to the unit circle but outside it for stationarity. These processes often create spiral/loop-like delay geometry.

(iii) **AR(3) and AR(4):** generate stable coefficients using roots or by scaling random coefficients conservatively.

(iv) **Near-lower-order cases:** set ϕ_p small. These test whether the method overclaims exact recovery.

(v) **Innovation distributions:** Gaussian, uniform, Student-t, and bounded uniform. Bounded innovations should make the slab/zonotope geometry most visible.

11.2 Evaluation metrics

For each method, report:

$$\Pr(\widehat{p} = p), \qquad \Pr(\widehat{p} \in \{p, p+1\}), \qquad \mathbb{E}|\widehat{p} - p|,$$

over Monte Carlo replications. Also report stability under sample size T, innovation variance, coefficient strength, and $k_{\max}$.

11.3 Minimal pseudocode

```
for replication in 1:R:
    simulate AR(p) sample X_1,...,X_T
    for k in 1:k_max:
        P_k = sliding_window_cloud(X, k)
        for epsilon in epsilon_grid:
            K[k, epsilon] = VR_complex(P_k, epsilon)
            c[k, epsilon] = functional(K[k, epsilon])
        Sigma_eps[k] = meet_matrix(c[k, :])
        v_eps[k] = principal_eigenvector(Sigma_eps[k])
        epsilon_crit[k] = epsilon_grid[argmax(v_eps[k])]
        summary[k] = c[k, epsilon_crit[k]]
    Sigma_k = similarity_matrix(summary[1:k_max])
    v_k = principal_eigenvector(Sigma_k)
    k_crit = argmax(v_k)
    p_hat = k_crit - 1   # if k is delay dimension
```

12 How to present precedence in the thesis

The literature can be organized as follows.

- **Classical time series:** Brockwell and Davis for ARMA definitions, stationarity, causal representations, prediction, and order selection [3].
- **Delay embeddings:** Takens for the original embedding theorem [12]; Sauer, Yorke, and Casdagli for embedology [11]; Abarbanel et al. for practical analysis of observed chaotic data [2].
- **TDA for time series:** Perea and Harer for sliding windows and persistence [9]; Karan and Kaygun for time-series classification via TDA [8].
- **Non-nested spaces:** Carlsson and de Silva for zigzag persistence [5].
- **Diagram comparison:** Reininghaus et al. for the persistence scale-space kernel [10]; Carriere et al. for sliced Wasserstein kernels [6]; Bubenik for persistence landscapes [4]; Adams et al. for persistence images [1].
- **Directed temporal structures:** Grigor'yan, Lin, Muranov, and Yau for path homology of digraphs [7].

13 Suggested thesis text for the key open problems

13.1 Open problem 1: lifting the dimension

The sequence $(\mathcal{P}_k)$ does not form a filtration in the usual TDA sense because the point clouds live in different ambient spaces. A lift

$$L_k : \mathbb{R}^k \to \mathbb{R}^{k_{\max}}$$

turns them into comparable clouds

$$\mathcal{Q}_k = L_k(\mathcal{P}_k) \subset \mathbb{R}^{k_{\max}}.$$

The lift should not be arbitrary. For AR processes, a statistically natural choice is the conditional-mean lift, where unobserved older lags are filled in by their best linear predictors from the observed k coordinates. This lift is model-compatible: for Gaussian AR processes, it coincides with conditional expectation, and for general stationary processes it remains the best linear predictor. Therefore changes in the lifted geometry as k increases can be interpreted as changes in available memory.

13.2 Open problem 2: constructing complexes without losing time

A pure Vietoris–Rips complex forgets the temporal order of the observations. A pure temporal complex forgets the observed values. Therefore the natural compromise is a temporal-geometric complex:

$$K_{k,\varepsilon} = T_k \cap \mathrm{VR}_\varepsilon(\mathcal{Q}_k),$$

where T_k encodes allowed consecutive time blocks and $\mathrm{VR}_\varepsilon(\mathcal{Q}_k)$ encodes geometric closeness. This construction preserves both pieces of information. The temporal part ensures that simplices represent local pieces of the trajectory; the geometric part ensures that these pieces are data-dependent.

13.3 Open problem 3: choosing the operation $\oplus$

If the complexes are nested, the meet/min construction is natural. If they are not nested, the union operation

$$\langle K, L \rangle_f = f(K) + f(L) - f(K \cup L)$$

measures shared complexity whenever f is subadditive. For persistence diagrams, it is more natural to skip an explicit complex operation and use a diagram distance or kernel. Thus there are two levels of operation: complex-level operations for common vertex sets, and diagram-level kernels for non-nested or non-comparable complexes.

13.4 Open problem 4: choosing f

The functional should produce a non-monotone or transition-sensitive profile in k. Pure face counts are usually monotone and therefore unsuitable alone. Good candidates are products of increasing and decreasing quantities, residual-weighted topological scores, persistence summaries, and intrinsic-dimension contrasts. In the AR setting, prediction residuals are especially important because the definition of AR order is itself a prediction-memory property.

14 Conclusion

Your idea is promising, but the cleanest version is not simply "replace ε by k." The embedding dimension k is not a filtration scale in the same way as a Vietoris–Rips radius. Instead, k controls how much temporal memory is represented. The method becomes mathematically meaningful when the spectral scale-selection principle is applied to a k-indexed profile that reflects geometry, prediction, persistence, or residual structure.

The most promising final thesis narrative is:

> We use TDA to convert each candidate memory length k into a topological-geometric summary. The spectral scale-selection principle then regularizes and ranks these summaries. For AR(p) processes, the summaries are expected to change until the correct memory length is reached and stabilize afterwards. Thus the selected dimension estimates the AR order, up to the natural p versus $p+1$ convention of delay embeddings.

This is ambitious but realistic. It gives you a defensible mathematical story, several implementable experiments, and a clear distinction between established precedence and your own novel contribution.

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